

Retail Sales Data Analysis

Task 1 – EDA on Retail Sales Data

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Track: Data Analytics

Table of Contents

1. Introduction
2. Objectives
3. Dataset Description
4. Tools and Technologies Used
5. Data Preprocessing
6. Exploratory Data Analysis
7. Key Findings
8. Business Recommendations
9. Conclusion

1. Introduction

Retail businesses generate large volumes of transactional data every day. Analyzing this data helps organizations understand customer purchasing behavior, identify sales trends, evaluate product performance, and make informed business decisions.

The objective of this project is to perform Exploratory Data Analysis (EDA) on a retail sales dataset using Python. Various visualization techniques have been employed to identify meaningful patterns, trends, and relationships within the data. The insights obtained from this analysis can support business strategies related to inventory management, customer segmentation, and revenue optimization.

2. Objectives

The primary objectives of this project are:

- Analyze monthly and quarterly sales trends.
- Study customer demographics based on age and gender.
- Identify the best-performing product categories.
- Compare revenue generated by different product categories.
- Explore relationships among numerical variables using correlation analysis.
- Generate actionable business recommendations based on analytical findings.

3. Dataset Description

The dataset contains retail transaction records consisting of customer information, product details, and sales information.

Attribute	Description
Transaction ID	Unique transaction identifier
Date	Date of purchase
Customer ID	Unique customer identifier
Gender	Gender of customer
Age	Age of customer
Product Category	Purchased product category
Quantity	Number of items purchased
Price per Unit	Price of each item
Total Amount	Total transaction value

The dataset contains three product categories:

- Beauty
- Clothing
- Electronics

4. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

5. Data Preprocessing

Before performing the analysis, the following preprocessing steps were carried out:

- Imported the dataset into Python.
- Converted the Date column into datetime format.
- Created new columns for Year-Month and Quarter to analyze sales trends.
- Generated an Age Group column from the Age attribute.
- Verified the dataset for missing values and inconsistencies.
- Selected numerical variables for correlation analysis.

6. Exploratory Data Analysis

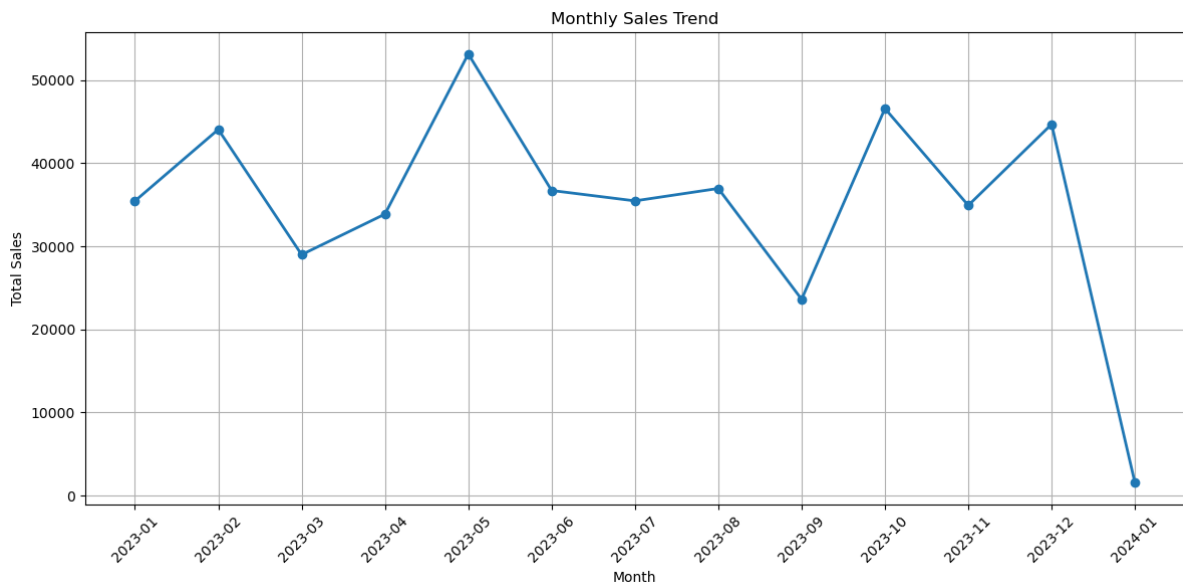
The following analyses were performed.

6.1 Monthly Sales Trend

A line chart was created to visualize monthly sales performance.

Observation

- The highest monthly sales were recorded in **May 2023**.
- The lowest monthly sales occurred in **January 2024**.

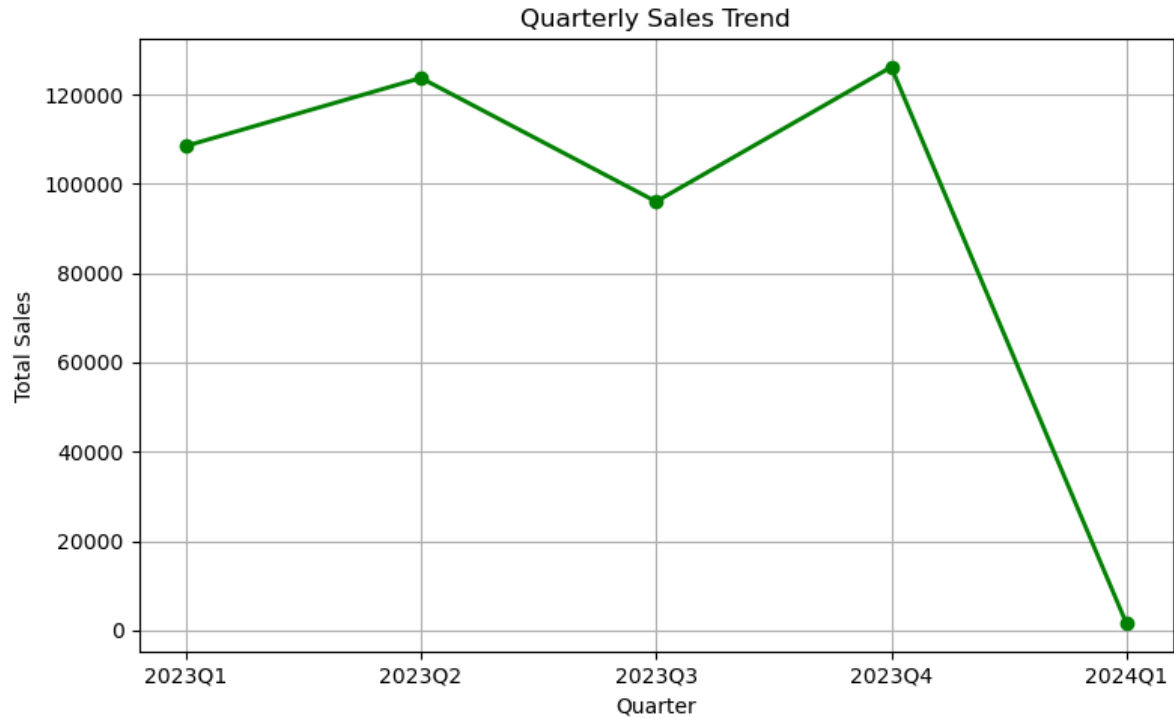


6.2 Quarterly Sales Trend

Quarterly sales were analyzed to understand seasonal business performance.

Observation

- The highest quarterly sales were recorded during **Q4 of 2023**.
- The lowest quarterly sales were observed during **Q1 of 2024**.



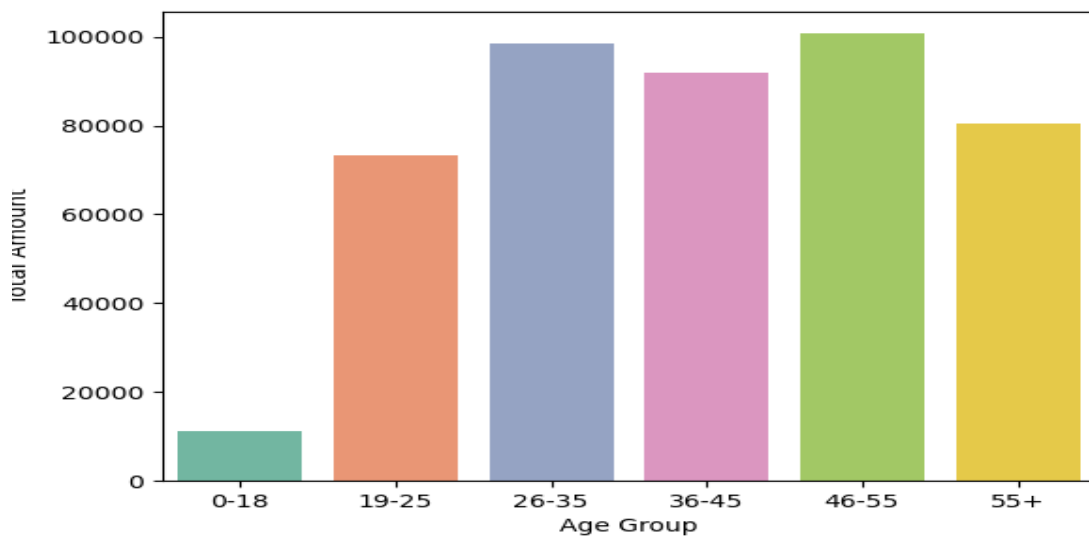
6.3 Customer Demographics

Customer distribution was analyzed based on age groups and gender.

Age Group Analysis

Observation

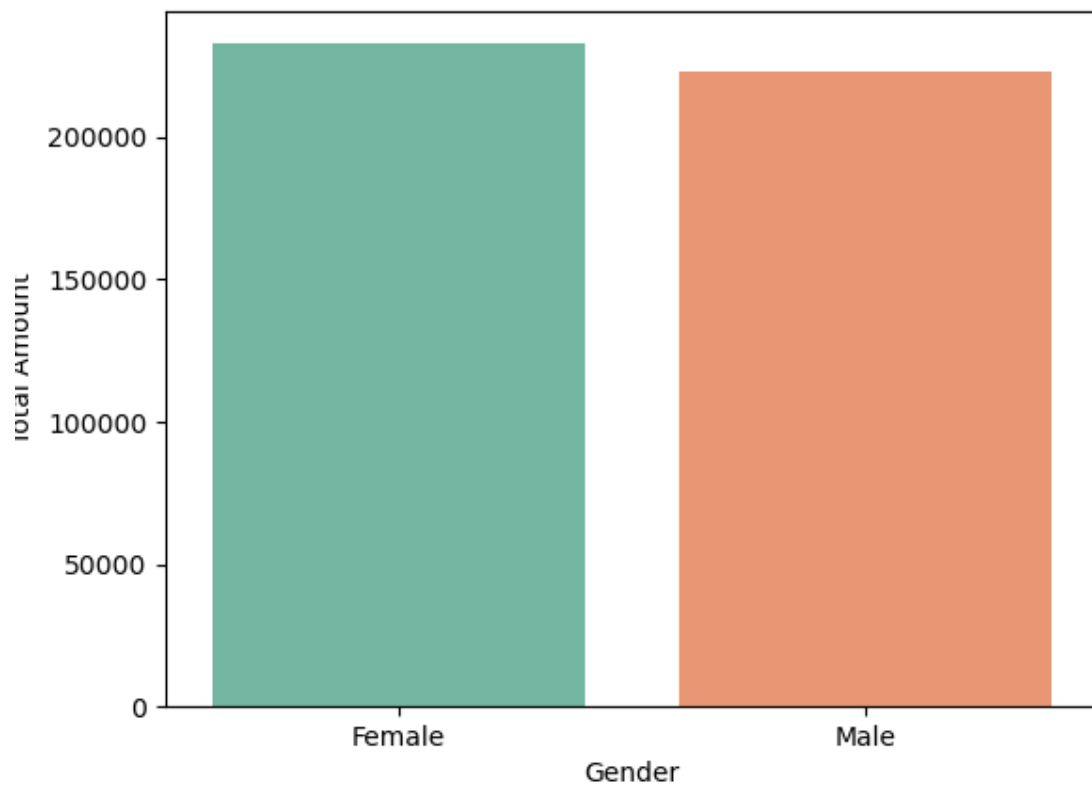
- Customers belonging to the **46–55** age group represent the largest customer segment.
- This age group also contributes the highest share of total sales.



Gender Analysis

Observation

- Female customers contribute more to total revenue compared to male customers.



6.4 Product Category Analysis

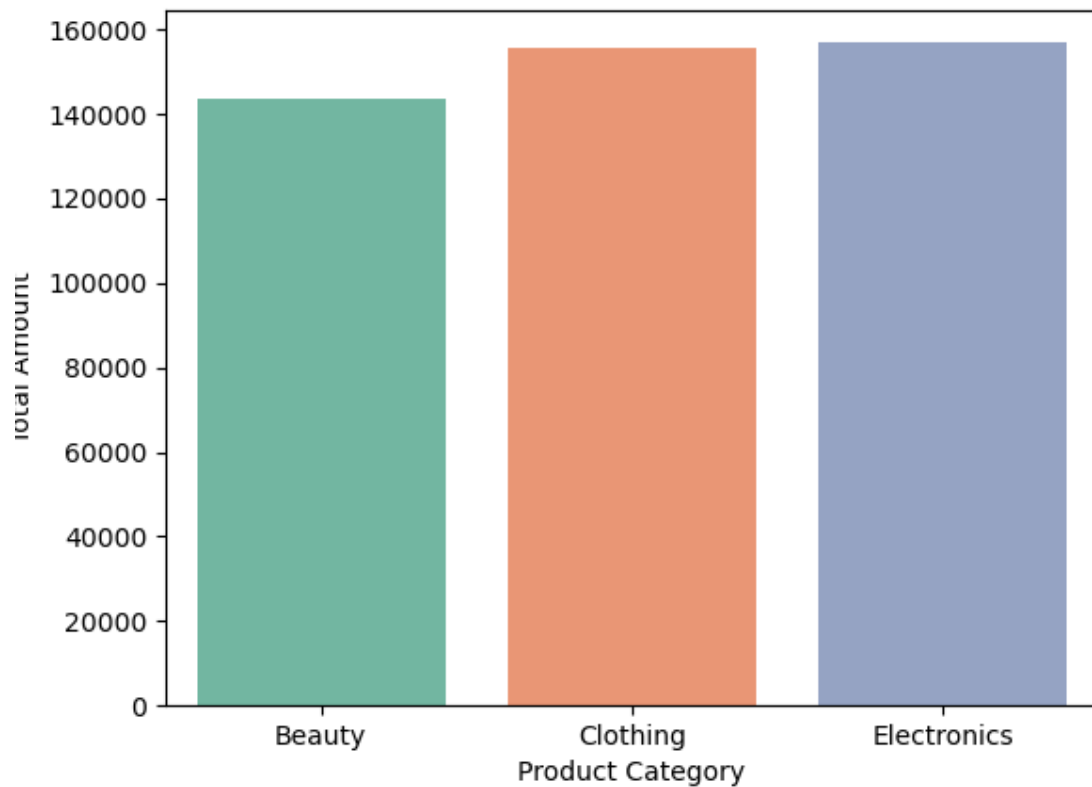
Product category performance was analyzed based on quantity sold and total revenue.

Observation

- Clothing records the highest quantity of products sold.
- Electronics generates the highest revenue.

Note

The dataset contains only three product categories and does not provide individual product names or product IDs. Therefore, identifying the top ten best-selling products was not possible

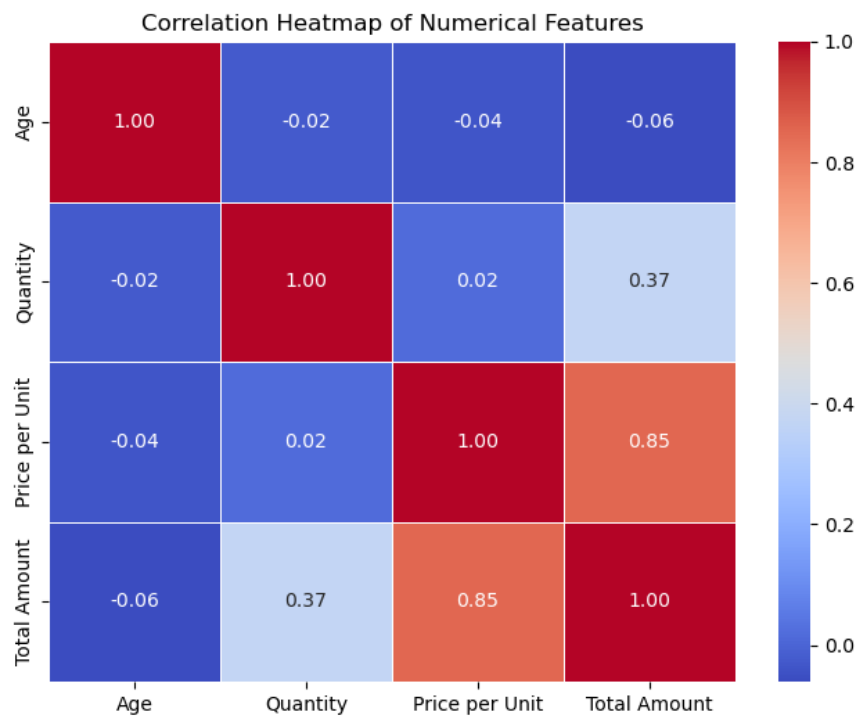


6.5 Correlation Analysis

A correlation heatmap was generated using the numerical variables.

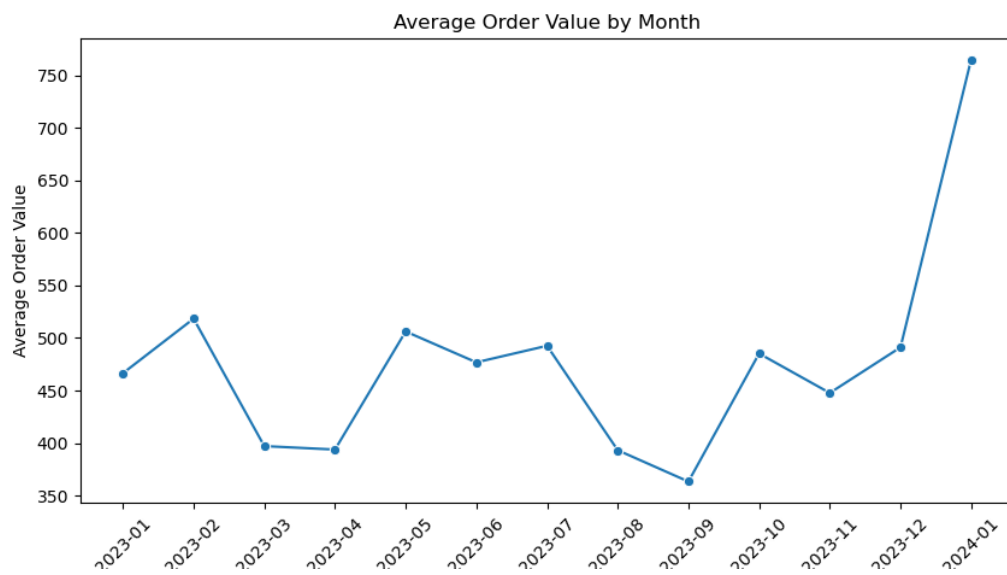
Observation

- Total Amount has a strong positive correlation with Quantity and Price per Unit.
- Age has very little correlation with sales-related variables.
- Quantity and Price per Unit show only a weak relationship.



6.6 Additional Analysis

An additional visualization was performed to gain deeper insights into customer purchasing behavior.



7. Key Findings

The analysis revealed the following important insights:

- Sales reached their highest-level during May 2023.
- Q4 of 2023 generated the maximum quarterly revenue.
- Customers aged 46–55 represent the most valuable customer segment.
- Female customers contribute more to overall sales.
- Electronics is the highest revenue-generating product category.
- Clothing has the highest sales volume.
- Total transaction value depends primarily on quantity purchased and product price.
- Customer age does not significantly influence transaction value.

8. Business Recommendations

Recommendation 1

Develop targeted marketing campaigns and loyalty programs for customers aged **46–55**, as they contribute the highest share of total revenue.

Recommendation 2

Increase inventory planning and promotional efforts for the Electronics category, as it generates the highest revenue despite lower sales volume than Clothing

Recommendation 3

Prepare inventory and marketing campaigns ahead of peak sales periods, particularly **Q4**, while offering promotional discounts during lower-performing months to improve sales consistency.

Recommendation 4

Enhance future data collection by recording individual product names and product IDs. This would enable detailed product-level analysis, inventory optimization, and identification of top-selling products.

9. Conclusion

This project successfully explored retail sales data using Exploratory Data Analysis techniques. Multiple visualizations were developed to examine sales trends, customer demographics, product performance, and numerical relationships within the dataset.

The findings indicate that customers aged **46–55** and female shoppers contribute significantly to business revenue. Electronics emerged as the highest revenue-generating category, while Clothing recorded the highest sales volume. Seasonal trends suggest stronger sales performance during Q4, emphasizing the importance of strategic inventory management and timely promotional campaigns.

Overall, this analysis demonstrates how data-driven decision-making can help retail businesses better understand customer behavior, optimize operations, and improve overall business performance.